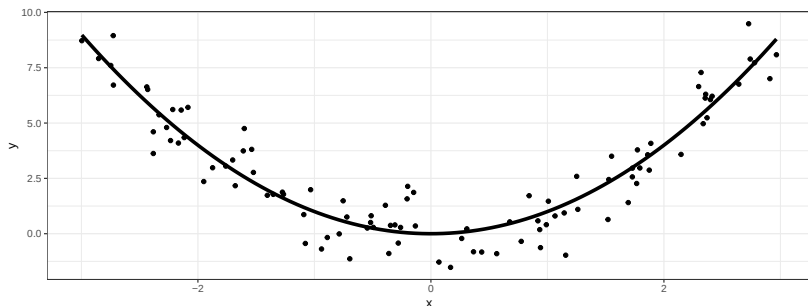


Ensembles and Random Forests

Ryan Miller

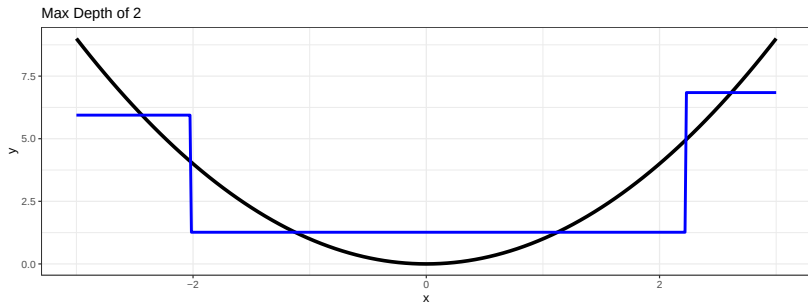
Introduction

Consider data where the true relationship is $Y = X^2 + \text{error}$:



What would the prediction line for a decision tree with a `max_depth` of 2 look like?

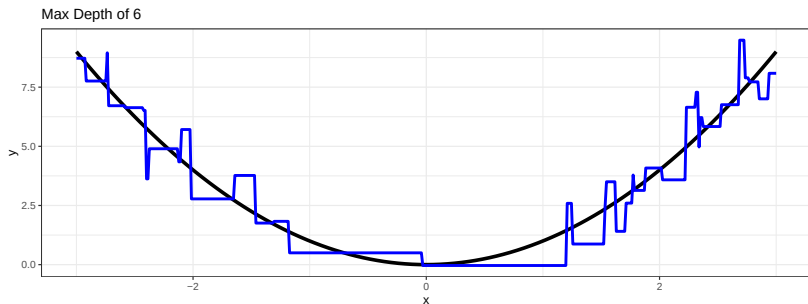
Introduction



Notice this model is highly biased, how could we get a decision tree model with lower bias?

Introduction

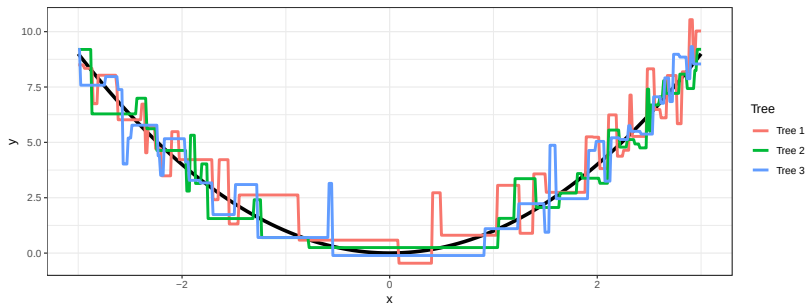
This model seems to pick up on the quadratic curve, but it exhibits high variance. . .



Let's next see how `max_depth=6` models differ from sample-to-sample

Introduction

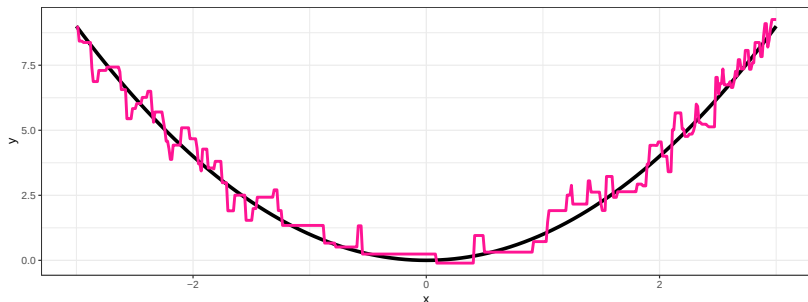
Below are three decision tree models all using `max_depth=6`, but trained on different samples of $n = 100$ using the quadratic data generation scheme we've been looking at:



How much variability is there in what these models will predict for $\hat{y}$ when $x = 1.5$?

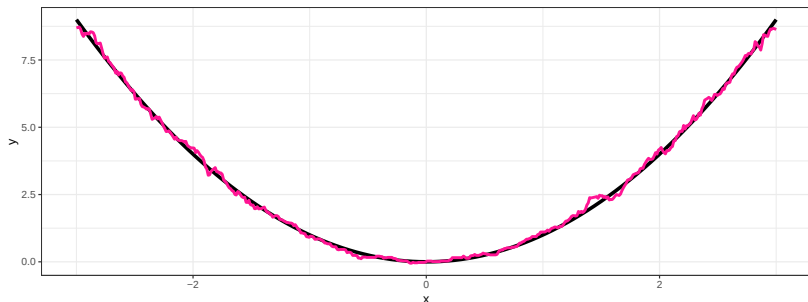
Averaging and Variability?

What if we averaged the predictions from the 3 different trees?



Averaging and Variability?

Now what if we scaled this up and used 50 different samples of size $n = 100$ and 50 different estimated trees?



Averaging many high-variance models can produce a lower-variance prediction.

Averaging and Variability?

A fundamental statistical principle is that averages exhibit less variability than individual values:

$$\text{Var}(\bar{X}) = \sigma^2/n < \text{Var}(X) = \sigma^2$$

when values are independent.

However, in practice we'll never have multiple independent samples to train different decision trees on, so how might we obtain multiple trees to average?

Bagging

Random forests rely upon *bagging*, or “bootstrap aggregation”, a procedure where each base learner (tree) is trained on a bootstrapped sample of training data:

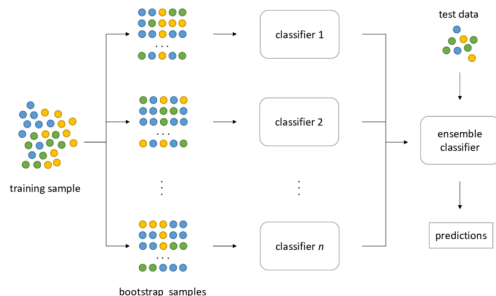


Image credit: <https://hudsonthames.org/bagging-in-financial-machine-learning-sequential-bootstrapping-python/>

Random Forest

- ▶ Bagging allows for some diversity among the base learners used in a random forest
 - ▶ However, because bootstrap samples are similar, the base learners will still be very similar unless other steps are taken
- ▶ Thus, the random forest algorithm also relies upon **feature subsampling** to further enhance the diversity of the base learners
 - ▶ Subsampling can be done at the level of the tree, within specific depths, or at individual nodes (splits)
 - ▶ Subsampling randomly selects a set of features and the decision tree splits within that level use only those features
- ▶ Combining bagging and subsampling makes the trees less correlated with one another, thereby allowing the random forest to benefit from the “wisdom of the crowd”

Random Forest Hyperparameters

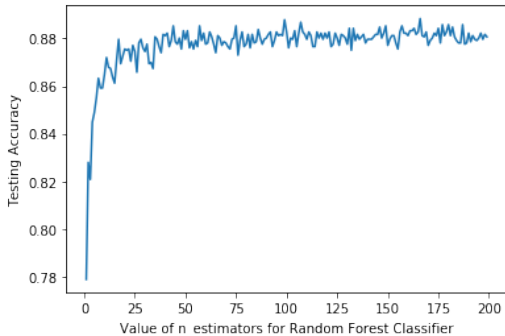
The performance of a random forest model depends upon hyperparameters that impact two different aspects of the algorithm:

1. Behavior of the base learners:
 - ▶ `max_depth`
 - ▶ `min_samples_split`
 - ▶ `min_impurity_decrease`
2. Number and diversity of base learners:
 - ▶ `n_estimators`
 - ▶ `max_features`

Note that random forests typically use deep trees. Rather than reducing variance by restricting depth, random forests reduce variance via bagging.

Random Forest Hyperparameters (cont.)

The `n_estimators` parameter is unique, as performance stabilizes once enough base learners are included in the forest, and more adding estimators *does not* lead to overfitting:



Ensemble Models

- ▶ An **ensemble model** is one that combines the predictions of multiple individual models known as **base learners**, usually via averaging (or majority voting for classification tasks)
- ▶ Random forests are an example of an ensemble model where decision trees are the base learners, but in practice an ensemble model could include any combination of base learners (with or without bagging)

What to Know for the Next Quiz

- ▶ Definition of an ensemble model, and the core steps of the random forest algorithm:
 - ▶ Create bootstrapped samples from the training data
 - ▶ Train a decision tree using a subsample of features on each bootstrapped sample
 - ▶ Get final predictions by aggregating the predictions of the individual decision trees (base learners)
- ▶ Have a basic understanding of the role of each major hyperparameter (`max_depth`, `n_estimators`, and `max_features` in particular)